EXTREMAL GRAPH THEORY

Erdős Problem 915

Maximizing Edge Density Subject to Connectivity Constraints

One-hour version

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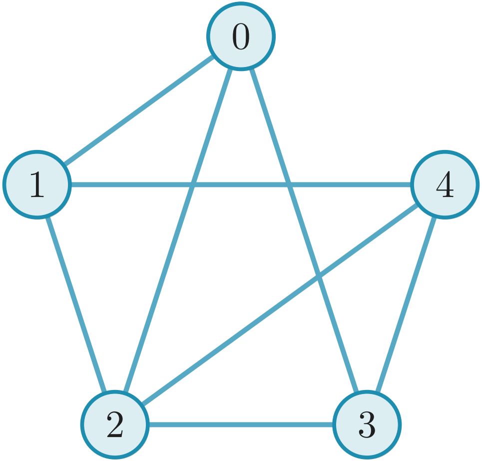
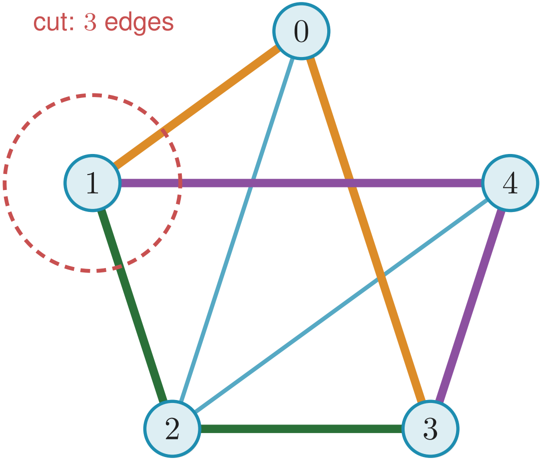
The graph $K_5 - \{04, 13\}$



The simple graph $K_5 - \{04, 13\}$ has five vertices and eight edges, where K_n is the complete graph on n vertices.

Menger's theorem

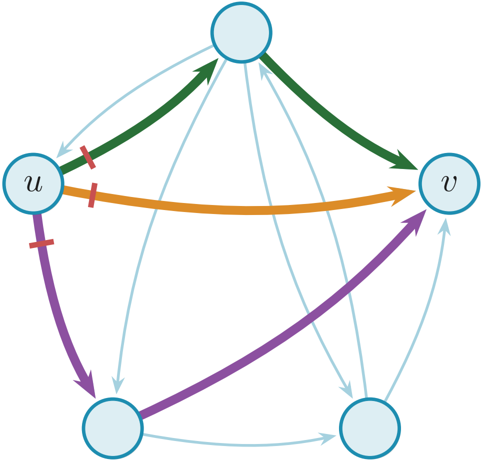


Menger's theorem on $K_5 - \{04, 13\}$: three edge-disjoint 1–3 routes (coloured) meet the minimal three-edge cut around vertex 1, so $\lambda(1, 3) = 3$.

Directed connectivity

$d^+(u) = 3$

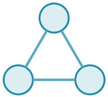
cut: 3 arcs



$d^-(v) = 4$

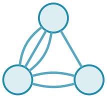
Three arc-disjoint $u \to v$ routes (coloured) and the matching three-arc directed cut (red) in a simple digraph. Thus $\lambda(u, v) = 3$, meeting the degree bound $\min(d^+(u), d^-(v)) = 3$.

The eight models



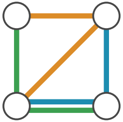
simple graph

one edge per pair



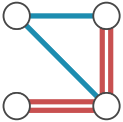
multigraph

parallel edges $\mu(u,v) \geq 1$



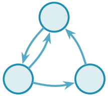
hypergraph

hyperedges span $r \geq 3$



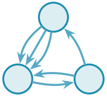
multihypergraph

a hyperedge repeated



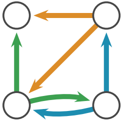
simple digraph

one arc per ordered pair



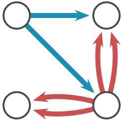
directed multigraph

parallel and opposite arcs



directed hypergraph

two share an edge

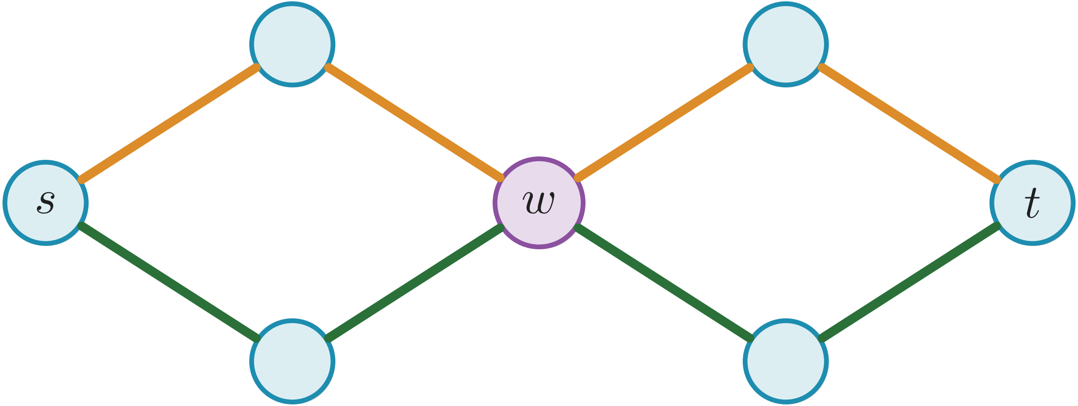


directed multihypergraph

a hyperedge repeated

The eight models side by side, undirected on top and directed below. The simple graph is a triangle, the multigraph has parallel edges, the hypergraph has three hyperedges, and the multihypergraph has one repeated hyperedge. The directed versions have arcs instead of edges, with the same pattern of repetition and hyperedges.

The separation axis



Two edge-disjoint s–t routes both pass through w, so $\lambda(s,t) = 2$ but $\kappa(s,t) = 1$.

Notation for the extremal functions

Model	edge-disjoint routes	internally vertex-disjoint routes
Simple graph	$\ell_m(n)$	$k_m(n)$
Multigraph	$L_m(n)$	$K_m(n)$
r-uniform simple hypergraph	$\ell_m^{(r)}(n)$	$k_m^{(r)}(n)$
r-uniform multihypergraph	$L_m^{(r)}(n)$	$K_m^{(r)}(n)$
Simple digraph	$\ell_m^{\mathrm{dir}}(n)$	$k_m^{\mathrm{dir}}(n)$
Directed multigraph	$L_m^{\mathrm{dir}}(n)$	$K_m^{\mathrm{dir}}(n)$
Directed r-uniform simple hypergraph	$\ell_m^{\mathrm{dir}\,(r)}(n)$	$k_m^{\mathrm{dir}\,(r)}(n)$
Directed r-uniform multihypergraph	$L_m^{\mathrm{dir}\,(r)}(n)$	$K_m^{\mathrm{dir}\,(r)}(n)$

Undirected bounds: multigraphs and Leonard

Theorem

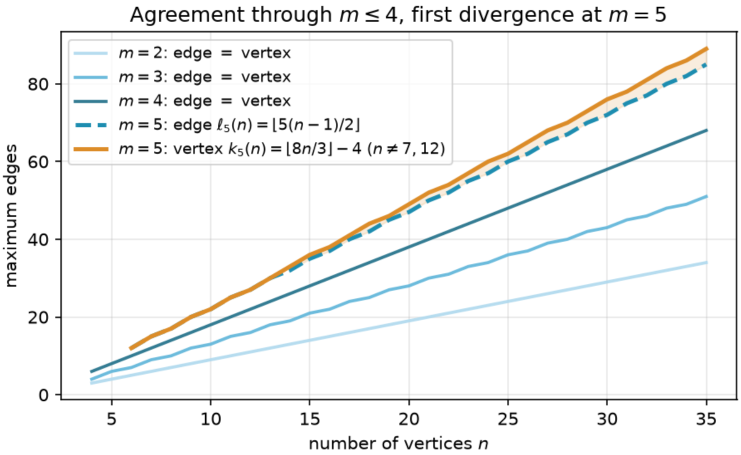
For undirected multigraphs on n vertices, $L_m(n) = (m-1)(n-1)$.

Theorem (Bártfai, Bollobás, Leonard)

For all $n \geq m$ and $m \leq 4$,

$$k_m(n) = \ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

Edge and vertex diverge



The edge and vertex extremal functions for $m = 2, 3, 4$ (blue shades, single curve per m since they agree) and $m = 5$ (dashed blue for edge, solid orange for vertex). The vertex function diverges from the edge function at $n = 14$ and grows faster thereafter.

Directed values at $m = 2$

Theorem (Directed arc value at $m = 2$, [AI])

For simple digraphs,

$$\ell_2^{\mathrm{dir}}(n) = \max\left(2(n-1),\ \left\lfloor \frac{n^2}{4} \right\rfloor\right).$$

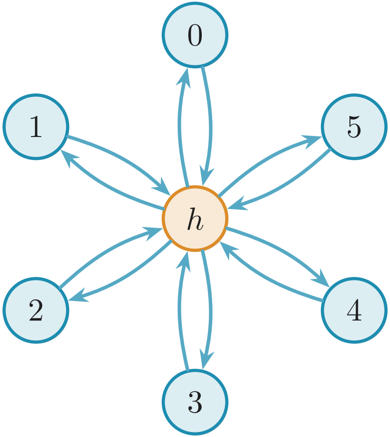
Theorem (Directed vertex value at $m = 2$, [AI])

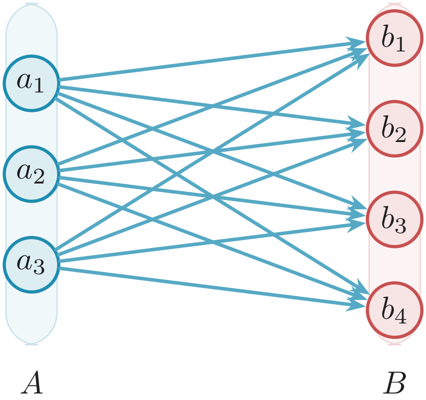
For simple digraphs, $k_2^{\mathrm{dir}}(n) = \ell_2^{\mathrm{dir}}(n)$*.*

Proposition (Hypergraph edge bound, [AI])

An r-uniform multihypergraph on n vertices with Berge edge-connectivity at most $m-1$ has total hyperedge multiplicity at most $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$. A simple hypergraph attains the bound whenever $m-1 \le \binom{n-2}{r-2}$, and a multihypergraph attains it whenever $(r-1) \mid n-1$.

The two branches tie at $n = 7$





The two branches tie at $n = 7$, where each holds 12 arcs at $\lambda^{\max} = 1$. Left: the bidirected star, a hub joined to six spokes both ways, $2(n-1) = 12$. Right: the one-directional complete bipartite digraph $B_{3,4}$, every arc from A to B, $\lfloor n^2/4 \rfloor = 12$.

Hypergraph vertices at $m = 2$

Theorem (Hypergraph vertex value at $m = 2$, [AI])

For $r \geq 2$ and $n \geq 1$, the maximum number of hyperedges of an r-uniform multihypergraph on n vertices with $\kappa^{\max} \leq 1$ is

$$K_2^{(r)}(n) \;=\; \left\lfloor \frac{n-1}{r-1} \right\rfloor,$$

attained by the star hypertree, which is simple. In particular $K_2^{(r)}(n) = k_2^{(r)}(n) = \ell_2^{(r)}(n)$: the vertex and edge problems agree at $m = 2$ for every uniformity, and repeated hyperedges never help.

Hypergraph vertices at $m = 3$

Theorem (Hypergraph vertex value at $m = 3$, [AI])

For $r \geq 2$, every r-uniform multihypergraph on n vertices with $\kappa^{\max} \leq 2$ has at most $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ hyperedges. For $r \geq 3$ with $2 \leq \binom{n-2}{r-2}$ the bound is attained by a simple hypergraph, so

$$k_3^{(r)}(n) = \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor = \ell_3^{(r)}(n).$$

A multihypergraph attains it whenever $(r-1) \mid (n-1)$ instead, by repeating one hyperedge rather than needing $\binom{n-2}{r-2}$ distinct ones, so $K_3^{(r)}(n)$ takes the same value on that range.

The quadratic bound: directed arcs and vertices

Theorem (The quadratic bound with a linear error, [AI])

For all $m \geq 2$ and $n \geq 2$, every simple digraph D on n vertices with $\lambda_D^{\max} \leq m-1$ satisfies

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4(m-1)(n-1),$$

so for each fixed m,

$$\ell_m^{\mathrm{dir}}(n) \;=\; \frac{n^2}{4} \;+\; \Theta_m(n) \qquad (m \geq 3),$$

and, for $m \geq 3$, both the leading constant $\frac{1}{4}$ and the order of the second-order term hold unconditionally. At $m = 2$, $\ell_2^{\mathrm{dir}}(n) = n^2/4 + O(1)$ as $n \to \infty$.

Theorem (Directed vertex problem, leading constant and order, [AI])

The same bound holds under the weaker hypothesis $\kappa_D^{\max} \leq m-1$, and consequently, for each fixed m,

$$k_m^{\mathrm{dir}}(n) \;=\; \frac{n^2}{4} \;+\; \Theta_m(n) \qquad (m \geq 3).$$

Directed multigraphs, closed

Theorem (Directed multigraph value, every n and m, [AI])

For all $n \geq 2$ and $m \geq 2$,

$$L_m^{\mathrm{dir}}(n) \;=\; (m-1)\, M(n), \qquad M(n) := \max\Big(2(n-1),\ \Big\lfloor \frac{n^2}{4} \Big\rfloor\Big).$$

The directed hypergraph constant

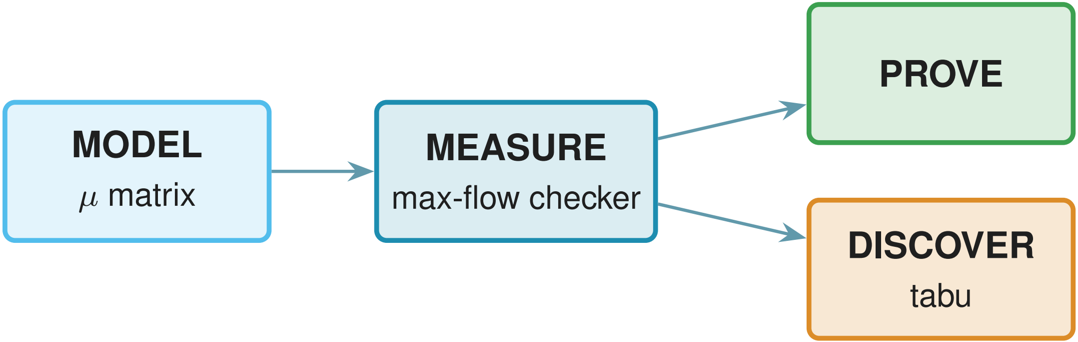
Theorem (The directed hypergraph constant, [AI])

Let $r \geq 2$ and $m \geq 2$. Every forward directed r-uniform multihypergraph $\mathcal{H}$ on $n \geq 2$ vertices with $\kappa_{\mathcal{H}}^{\max} \leq m-1$ satisfies

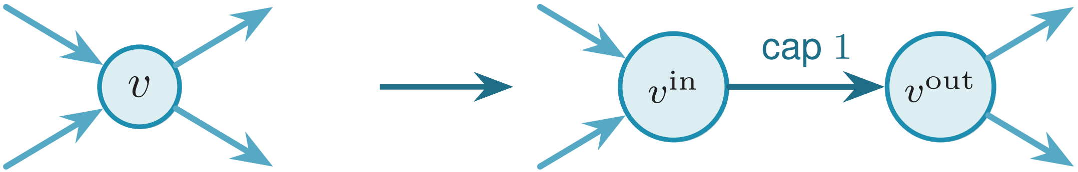
$$|E(\mathcal{H})| \;\leq\; \frac{m-1}{r-1}\left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4(m-1)^2(n-1).$$

The same bound therefore holds under $\lambda_{\mathcal{H}}^{\max} \leq m-1$, which is the stronger hypothesis by $\kappa \leq \lambda$. Hence for fixed r and m both the edge- and the vertex-disjoint maxima are $\big(1+o(1)\big)(m-1)n^2/\big(4(r-1)\big)$, and the bipartite construction is asymptotically optimal for both.

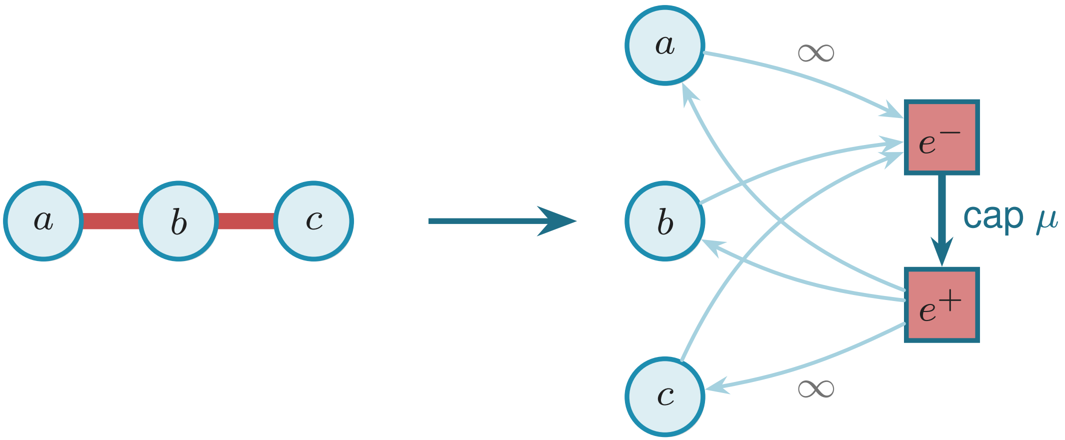
The architecture



Transformations



Transformations



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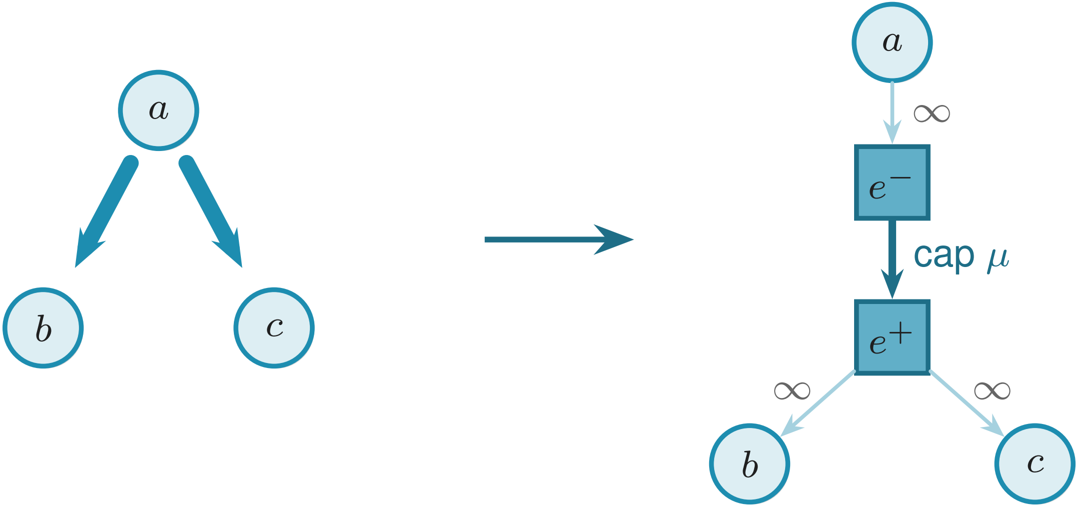
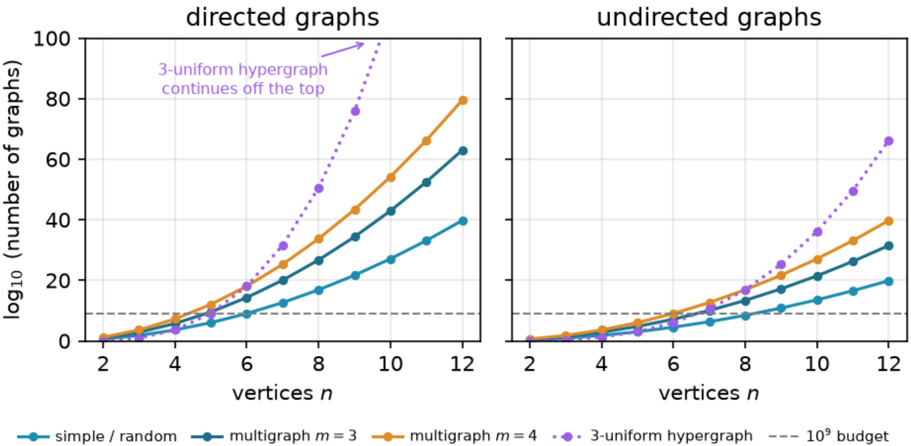
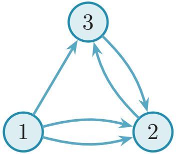


Candidate counts



The two encodings

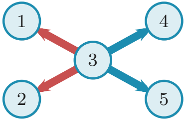


directed multigraph G

$$\xrightarrow{\mu} \begin{pmatrix} \mathbf{0} & 2 & 1 \\ 0 & \mathbf{0} & 1 \\ 0 & 1 & \mathbf{0} \end{pmatrix}$$

multiplicity matrix μ

Diag(μ)$= 0$ (no loops)



directed hypergraph H

$\longrightarrow$

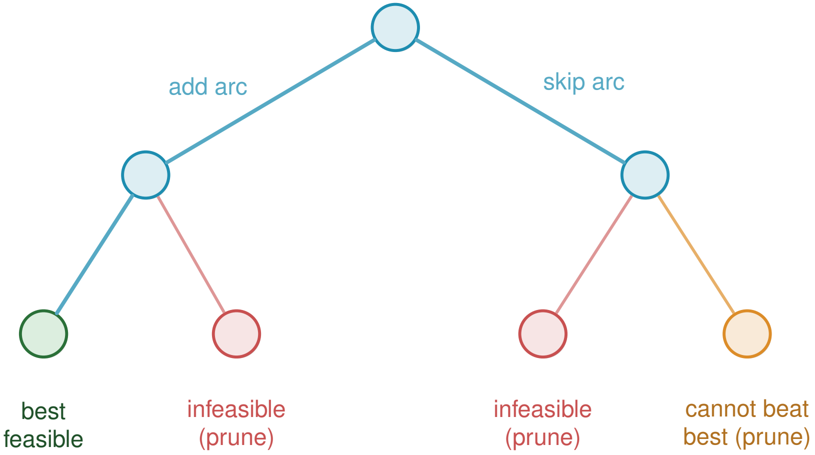
list of hyperedges:

$e_1 : \{3 \mid 1, 2\}$

$e_2 : \{3 \mid 4, 5\}$

The two encodings, for the directed variants. The digraph is the multiplicity matrix μ, where μ_{ij} counts arcs from i to j. The directed hypergraph is a list of hyperedges, with the tail of each isolated. Multihypergraphs repeat at least one entry in the list.

Branch and bound

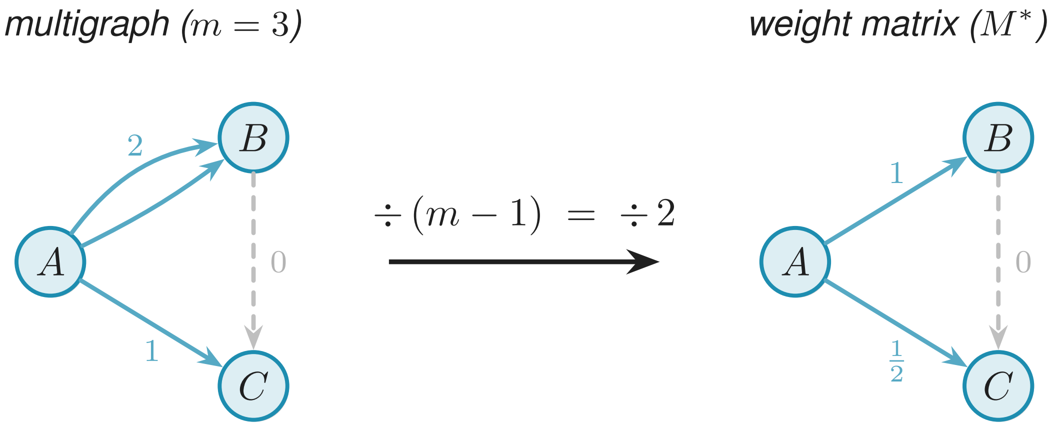


Each level of the partially built graph fixes one adjacency. Infeasible partial graphs (red) and branches unable to beat the incumbent (orange) are discarded.

Generating search: a benchmark

n	$\ell_2^{\mathrm{dir}}(n)$	blind (s)	pruned (s)	generated (s)
3	4	0.05	0.003	0.009
4	6	6.3	0.008	0.008
5	8	–	0.22	0.039
6	10	–	14.5	0.73
7	12	–	1156.6	17.0

Scaling out m



Scaling at $m = 3$: dividing multiplicities by $m - 1 = 2$ turns the integer multigraph into a weight matrix with entries in $[0, 1]$ and pairwise maximum flow at most one. Hence one m-free optimisation bounds every threshold.

The search: an energy function

To implement the metaheuristic, give each graph one number: its energy

$$\Phi(G) = -|E(G)| + \Lambda \cdot \max\big(0,\ c^{\max}(G) - (m-1)\big).$$

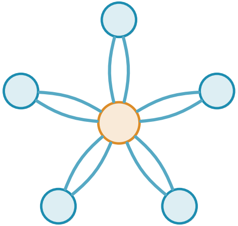
What the search rediscovers

Construction	Variant	n	m	Search	Theory
spanning tree (simple undirected)	$\ell_m(n)$	6	2	5	5
multiplicity-$(m-1)$ star (multi undirected)	$L_m(n)$	6	3	10	10
bidirected star (simple directed)	$\ell_m^{\mathrm{dir}}(n)$	4	2	6	6
bidirected star (simple directed)	$\ell_m^{\mathrm{dir}}(n)$	6	2	10	10
hub–bipartite tie (simple directed)	$\ell_m^{\mathrm{dir}}(n)$	7	2	12	12
double star (multi directed)	$L_m^{\mathrm{dir}}(n)$	4	3	12	12
double star (multi directed)	$L_m^{\mathrm{dir}}(n)$	5	3	16	16

Three rediscovered shapes

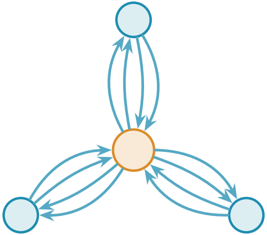
spanning tree

$n = 6$, $\ell_2(n) = 5$



multigraph star

$n = 6$, $m = 3$, $L_3(n) = 10$

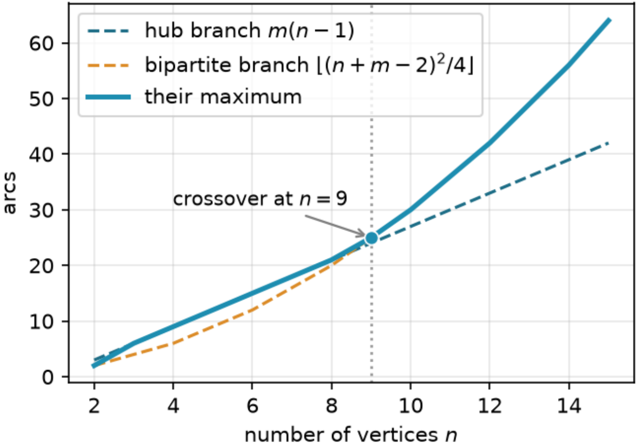


double star

$n = 4$, $m = 3$, $L_3^{\mathrm{dir}}(n) = 12$

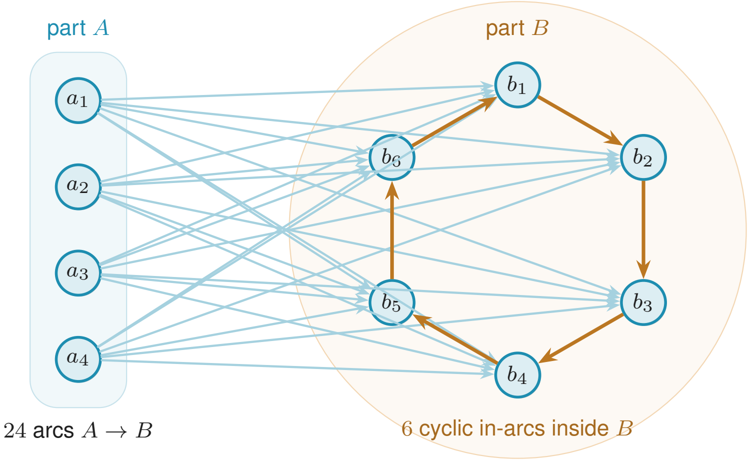
Three of the shapes the search rediscovers, each drawn at the size the table reports. The fourth, the bidirected star tying the one-directional wall at $n = 7$, was shown earlier.

The directed crossover



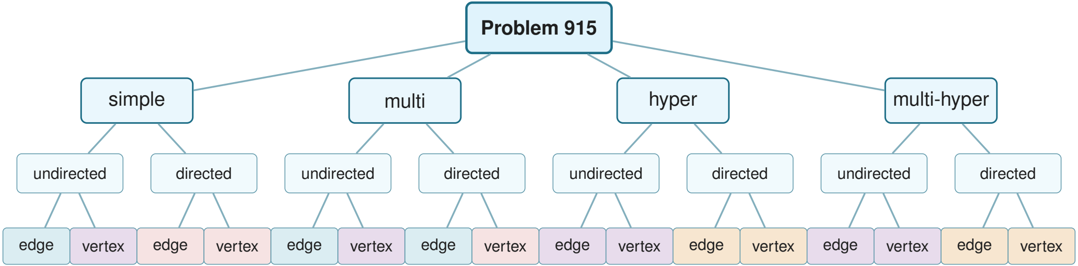
The two lower-bound branches of the directed-arc conjecture at $m = 3$, and the maximum they define.

The augmented bipartite digraph



The augmented bipartite digraph at $m = 3$, $n = 10$. Every cross-part arc runs from A to B, and inside B each vertex takes one arc from its cyclic predecessor, which is $m - 2$ arcs per vertex. A pair a_i, b_j then has the direct arc and one detour, so $\lambda^{\max} = 2$, and the count is $24 + 6 = 30$ against the hub branch's 27.

Status of the sixteen variants



closed form proved for every n and m

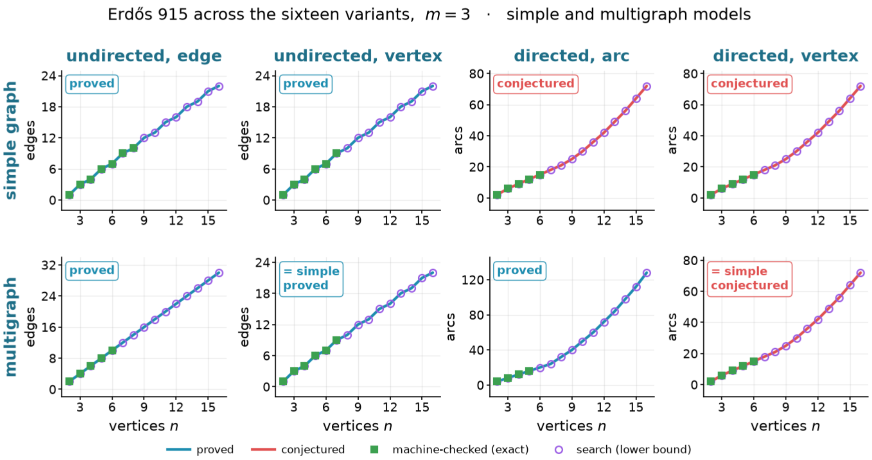
leading constant proved, exact value open

conjectured, matching certified small n

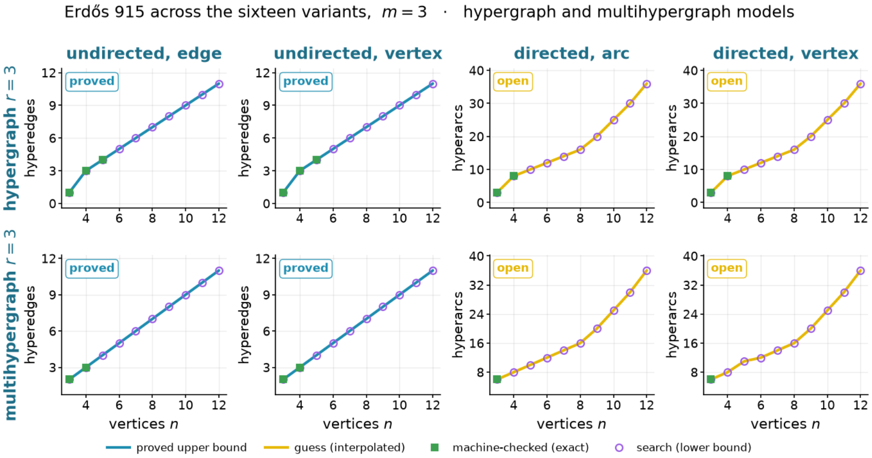
exact on stated parameter ranges, open outside them

Status of the sixteen variants. Colour marks the status of a leaf: blue a closed form proved for every n and m, purple an exact value proved on stated parameter ranges and open outside them, red a conjectured exact formula with a proved lower bound, and amber a proved leading term with the exact value unknown.

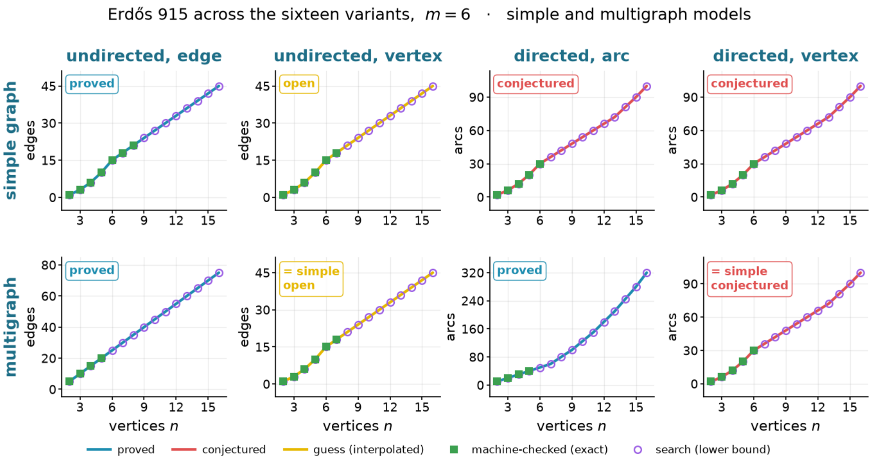
Eight graph variants at $m = 3$



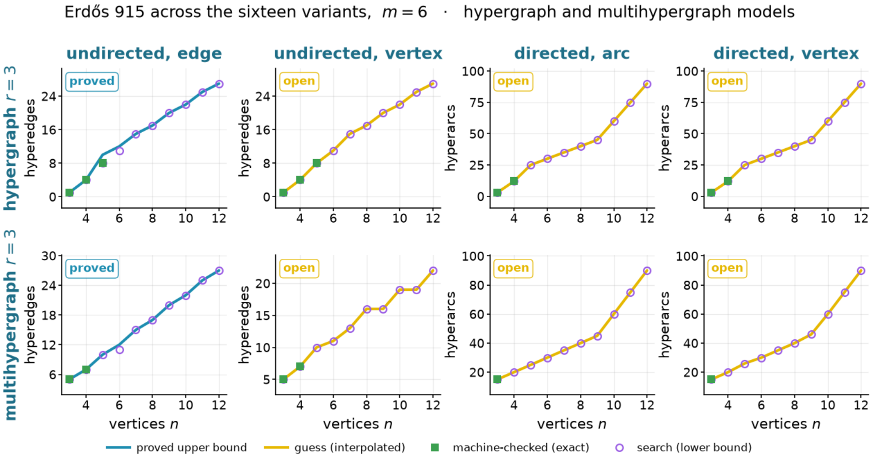
Eight hypergraph variants at $m = 3$



Eight graph variants at $m = 6$



Eight hypergraph variants at $m = 6$



The sixteen variants, summarised

Model	Edge or arc-disjoint	Vertex-disjoint
Undirected		
simple graph	$\lfloor m(n-1)/2 \rfloor$, proved	the edge value for $m \le 4$, and $\lfloor 8n/3 \rfloor - 4$ at $m = 5$; open for $m \ge 6$
multigraph	$(m-1)(n-1)$, proved	the simple value
r-hypergraph	$\le \lfloor (m-1)(n-1)/(r-1) \rfloor$, with equality in the range below	$\lfloor (n-1)/(r-1) \rfloor$ at $m = 2$ and $\le \lfloor 2(n-1)/(r-1) \rfloor$ at $m = 3$; open for $m \ge 4$
r-multi-hyper	the same bound, with equality when $(r-1) \mid (n-1)$	the same bounds, repeats closing cases the simple model misses at $m = 3$
Directed		
simple digraph	$M(n)$ at $m = 2$, proved; $n^2/4 + \Theta_m(n)$ for $m \ge 3$, exact value conjectured	$M(n)$ at $m = 2$, proved; $n^2/4 + \Theta_m(n)$ for $m \ge 3$, equality with the arc value open
multigraph	$(m-1)M(n)$, proved	the simple digraph value
r-hypergraph	$(1+o(1))(m-1)n^2/[4(r-1)]$ for fixed r, m, leading term proved	the same leading term for fixed r, m
r-multi-hyper	the same leading term, the bound proved with repeats allowed	the same leading term for fixed r, m

Exact values, $m = 3$

Model	Variant	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$	$n=8$
simple graph	undirected, edge ($\ell_m(n)$), proved	1	3	4	6	7	9	10
	undirected, vertex ($k_m(n)$), proved	1	3	4	6	7	9	10
	directed, arc ($\ell_m^{\mathrm{dir}}(n)$), conjectured	2	6	9	12	15	$\geq$18	$\geq$21
	directed, vertex ($k_m^{\mathrm{dir}}(n)$), conjectured	2	6	9	12	15	$\geq$18	$\geq$21
multigraph	undirected, edge ($L_m(n)$), proved	2	4	6	8	10	12	14
	undirected, vertex ($k_m(n)$), = simple, proved	1	3	4	6	7	9	10
	directed, arc ($L_m^{\mathrm{dir}}(n)$), proved	4	8	12	16	20	24	32
	directed, vertex ($k_m^{\mathrm{dir}}(n)$), = simple, conjectured	2	6	9	12	15	$\geq$18	$\geq$21
hypergraph $r = 3$	undirected, edge, proved		1	3	4	5	6	7
	undirected, vertex, proved		1	3	4	5	6	7
	directed, arc, open		3	8	$\geq$10	$\geq$12	$\geq$14	$\geq$16
	directed, vertex, open		3	8	$\geq$10	$\geq$12	$\geq$14	$\geq$16
multihypergraph $r = 3$	undirected, edge, proved		2	3	4	5	6	7
	undirected, vertex, proved		2	3	4	5	6	7
	directed, arc, open		6	$\geq$8	$\geq$10	$\geq$12	$\geq$14	$\geq$16
	directed, vertex, open		6	$\geq$8	$\geq$10	$\geq$12	$\geq$14	$\geq$16

Exact values, $m = 6$

Model	Variant	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$	$n=8$
simple graph	undirected, edge ($\ell_m(n)$), proved	1	3	6	10	15	18	21
	undirected, vertex ($k_m(n)$), open	1	3	6	10	15	18	$\geq$21
	directed, arc ($\ell_m^{\mathrm{dir}}(n)$), conjectured	2	6	12	20	30	$\geq$36	$\geq$42
	directed, vertex ($k_m^{\mathrm{dir}}(n)$), conjectured	2	6	12	20	30	$\geq$36	$\geq$42
multigraph	undirected, edge ($L_m(n)$), proved	5	10	15	20	25	30	35
	undirected, vertex ($k_m(n)$), = simple, open	1	3	6	10	15	18	$\geq$21
	directed, arc ($L_m^{\mathrm{dir}}(n)$), proved	10	20	30	40	50	60	80
	directed, vertex ($k_m^{\mathrm{dir}}(n)$), = simple, conjectured	2	6	12	20	30	$\geq$36	$\geq$42
hypergraph $r = 3$	undirected, edge, proved		1	4	8	$\leq$12	15	17
	undirected, vertex, open		1	4	8	$\geq$11	$\geq$15	$\geq$17
	directed, arc, open		3	12	$\geq$25	$\geq$30	$\geq$35	$\geq$40
	directed, vertex, open		3	12	$\geq$25	$\geq$30	$\geq$35	$\geq$40
multihypergraph $r = 3$	undirected, edge, proved		5	7	10	$\leq$12	15	17
	undirected, vertex, open		5	7	$\geq$10	$\geq$11	$\geq$13	$\geq$16
	directed, arc, open		15	$\geq$20	$\geq$25	$\geq$30	$\geq$35	$\geq$40
	directed, vertex, open		15	$\geq$20	$\geq$26	$\geq$30	$\geq$35	$\geq$40

What remains open

Problem	Known here	Remaining target
Directed simple arc, $m \geq 3$	$\ell_m^{\mathrm{dir}}(n) = n^2/4 + \Theta_m(n)$	Prove the decomposition, or otherwise determine the linear coefficient.
Directed simple vertex, $m \geq 3$	$k_m^{\mathrm{dir}}(n) = n^2/4 + \Theta_m(n)$	Decide whether its exact value equals the arc value, given that an arc upper bound does not transfer.
Directed multigraph classification	Exact value for all n, m and quadratic-branch uniqueness for $n \geq \max(8, 2m)$	Classify the range $n < 2m$ and the remaining linear-branch extremisers.
Undirected simple vertex, $m \geq 6$	Block formulation and $m/2 < c_m \leq 2(m-1)$, where $c_m = \lim_n k_m(n)/(n-1)$	Determine c_m, whose likely next structure is triconnected rather than merely blockwise.
Multigraph vertex, parallel copies distinct	Block-knapsack formulation and order $\Theta(m^2 n)$ as $m \to \infty$ with $n/m \to \infty$, with the per-vertex constant between $3 - 2\sqrt{2}$ and 2	Determine the asymptotic constant and the optimal 2-connected blocks.
Hypergraph vertex, $m \geq 4$	Exact at $m = 2$ and, in the stated range, $m = 3$	Locate the first failure for $r \geq 3$ and replace the rank-two decomposition used at $m = 3$.
Directed hypergraphs	For fixed r, m, leading term $(m-1)n^2/(4(r-1))$ for all orientations and both separations	Determine finite exact values and whether forward and general agree once $n > r$.

AI: curse or blessing?

Pros

- LaTeX typesetting
- length & complexity of generated text

Cons

- length & complexity of generated text
- control

It's a tool.

short leash $+$ AI proofs $=$ argumented conjecture [AI]